

DocScout: When Does Read-Budget Reinforcement Learning Help for Document-Search Agents?

A Diagnose-Fix-Verify Study: When (and Why) Read-Budget RL Pays Off

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Abstract

A retrieval agent that decides *how much to read* and *when to stop* should beat fixed top- k RAG: matching its accuracy while spending fewer tokens on irrelevant text. We test this for natural-language documents. **DocScout** is a tool-using agent (`search/read/expand/answer`) trained with a read-budget reward, in the spirit of CodeScout but with read-cost as a first-class objective. Our first design was silently inert: REINFORCE produced *zero* policy movement (0/150 held-out instances changed), and the agent was dominated by one-shot RAG. Rather than tune until it won, a diagnose-before-train protocol traced this to two fixable design flaws: (i) the reward was *flat at the SFT optimum* (a weak efficiency term the SFT already maximized), and (ii) the learning rate was too small to move a peaked-SFT policy. Fixing both—a reward with a term the SFT does *not* maximize (a per-read action cost, which creates gradient variance) plus a selection bonus (rewarding reading the gold section first), at an adequate LR—yields a strong *positive* result: read-budget RL **Pareto-dominates** both SFT and one-shot RAG on the accuracy-per-read-token frontier, reaching **0.84 accuracy at 170 committed read-tokens** (reading exactly one section) vs. SFT’s 0.65 and RAG- $k1$ ’s 0.753@182, and matching RAG- $k5$ ’s accuracy at **18.7%** of its read cost ($5.3\times$ fewer tokens). The gain is consistent across two seeds and decomposes cleanly: RL fixes selection (gold-hit 84% with one read, $>$ RAG’s 75%), extraction (accuracy given gold read: $0.72\rightarrow 1.0$), and efficiency (reads per question: $3.25\rightarrow 1.0$). We contribute (a) the positive frontier result, (b) the diagnose-fix-verify methodology that turned an inert setup into a working one, and (c) a mechanistic recipe (non-flat reward + adequate LR) for read-budget RL on peaked-SFT tool agents.

1 Introduction

Standard retrieval-augmented generation (RAG) [9] retrieves a fixed number of passages once and pastes them into the prompt. This is wasteful—irrelevant passages consume context—and gives the model no control over *how much* it reads. A natural fix is an *agent* that can call `search`, selectively read full sections, and `answer` when it has enough evidence, trained with reinforcement learning (RL) to optimize not just accuracy but *reading efficiency*. This is the premise of recent tool-using retrieval agents [3,6,8] and of CodeScout [11], which trains a code-localization agent with RL and a minimal tool set.

We ask the natural-language analogue: **can a read-budget RL agent dominate fixed- k RAG on the accuracy-per-read-token frontier?** We build DocScout, port CodeScout’s minimal-tool recipe to text documents, and make read-budget a first-class reward objective via an *efficiency-ratio* term (the fraction of read tokens that are gold evidence).

Our first design was silently inert—and we report the fix, not the burial. With a natural read-budget reward (efficiency-ratio), REINFORCE produced *zero* policy movement (0/150 held-out instances changed), leav-

ing the agent dominated by one-shot RAG. We did not tune until the method won; we diagnosed the failure, fixed it, and verified the fix:

- **Diagnose.** The reward was *flat at the SFT optimum*: SFT demonstrations already read the gold section, so the efficiency-ratio was already ≈ 1 and the reward could not distinguish better policies. Worse, the learning rate (10^{-6}) was too small for a LoRA update to change the peaked-SFT greedy decode at all.
- **Fix.** Add a reward term the SFT does *not* maximize—a per-read *action cost* that penalizes the over-reading SFT inherits, creating real gradient variance—plus a *selection bonus* that rewards reading the gold section first, and raise the LR to 10^{-5} .
- **Verify.** The fixed read-budget RL now *Pareto-dominates* SFT and one-shot RAG: 0.84 accuracy at 170 committed read-tokens (reading exactly one section) vs. SFT’s 0.65 and RAG- $k1$ ’s 0.753@182, matching RAG- $k5$ ’s accuracy at 18.7% of its read cost.

Contributions.

- A **positive frontier result**: read-budget RL Pareto-dominates SFT and fixed- k RAG on the accuracy-per-read-token frontier (0.84@170 BPE, $n_{\text{read}}=1$, consistent over 2 seeds and 6 checkpoints; §6).
- A **mechanistic recipe** for read-budget RL on peaked-SFT tool agents: the reward must contain a term the SFT does *not* maximize (action cost $\Rightarrow$ gradient variance) plus a selection objective, at an LR that can move the greedy policy (§7).
- A **diagnose-fix-verify methodology** that turned an inert setup into a working one, and that independently surfaced three integrity bugs in our own earlier results (App. B)—a low-cost guardrail for agentic-RL work.
- A **realistic-scale, leakage-free substrate** (synth-v3) and an honest BPE-token cost metric, where RAG genuinely desaturates (§4).
- Two **reusable findings**: the SFT stop-failure pathology and its $4\times$ answer-up-weighting fix, and REINFORCE-with-baseline’s superiority over GRPO on peaked-SFT tool agents (§5).

Why the journey matters. Reporting only the final positive result would hide the trap: a read-budget reward that *looks* reasonable is silently inert on a peaked-SFT policy, and an LR that is standard for full fine-tuning is too small for LoRA RL. By exposing the inert v1 and its fix, we make it cheaper for others to get read-budget RL to work—and harder to mistake “RL didn’t help” for “RL can’t help.”

2 Related Work

Tool-using retrieval agents. ReAct [15] and WebGPT [10] establish tool-using agents for QA but do not shape read cost with RL. FLARE [7] retrieves actively when generation confidence drops—a prompting-time heuristic for “when to retrieve,” orthogonal to our learned, reward-shaped reading depth. Search-R1 [8], ReSearch [6], AutoSearch [3] and dynamic-depth variants [2] train retrieval agents with RL, but they optimize answer accuracy and search *depth on the open web* (how many searches), not *intra-document reading volume* against a fixed corpus with section-locatable evidence. Their cost axis is #queries; ours is committed read tokens.

CodeScout and the minimal-tool recipe. CodeScout [11] is the closest analogue: a minimal-tool RL agent for code localization. Two key transfers guided our design: (i) a minimal tool set keeps the action space small and rollouts cheap (no execution/sandbox needed for localization, just as none is needed for reading); (ii)

CodeScout reports an efficiency *evaluation* (steps/tokens, their Tab. 6) but its training reward contains *no* efficiency term—efficiency is enforced only by a hard `max_turns` cutoff and a binary termination bonus r_{turn} (enabled only for the 14B model). DocScout’s efficiency-ratio reward is the natural-language, read-budget generalization: we make read-cost a *soft, first-class* reward term and study when it actually moves the policy.

Multi-hop QA and document-navigation baselines. MuSiQue [13] and HotpotQA [14] provide gold supporting paragraphs (section-locatable evidence) and are our generalization substrates. DeepRead [4] and IntraAgent [5] are prompting-based document agents without read-budget RL. ALDEN [1] is the strongest conceptual competitor: a VLM agent with an explicit *anti-redundancy* reading penalty. We implement its redundancy penalty as our R_{redund} variant; like our v1 efficiency-ratio, a pure reading-volume penalty is inert on a peaked-SFT policy, and our fix (a term the SFT does not maximize + selection bonus) is what makes read-budget RL move (§7).

GRPO / critic-free RL. GRPO [12] and its length-unbiased variant (DR.GRPO) underpin recent tool-agent training. We adopt the critic-free, no-KL, no-std-normalization recipe and show that, on peaked-SFT policies, both GRPO and a too-weak reward are silently inert—a failure mode we diagnose and fix (§6, §7).

Diagnostic methodology. Agentic-RL papers predominantly report final positive numbers. We failure-classify every run and explicitly report the inert starting point and its mechanistic fix, arguing this diagnose-fix-verify transparency is a needed complement to single-point positive claims.

3 Method

Environment as an MDP. A corpus of multi-section NL documents. Tools: `search(query)` returns top- k section *snippets* (first 40 words; cheap); `read(doc, section)` returns one full section (costly—the read-budget cost); `expand(doc, section, dir)` returns a neighbor; `answer(text, evidence)` terminates. Re-reading a section returns “already read” with zero new content (anti-redundancy guard). The environment is the authoritative accountant: it tracks which sections were read and how many *committed* tokens (READ/EXPAND content, not snippet glimpses) entered context. A three-tier compact memory (question + action log + last- N reads) is re-rendered each turn. Formally, state $s_t = (\text{question}, \text{action log}, \text{current candidates}, \text{recently-read sections}, \text{remaining budget})$; action $a_t \in \{\text{SEARCH}, \text{READ}, \text{EXPAND}, \text{ANSWER}\}$; the episode

ends on ANSWER or budget exhaustion.

Reward and anti-hacking design. Three variants, ablated as the core object:

$$\begin{aligned} R_{\text{add}} &= s_a + w \cdot F_1 - \lambda T_{\text{read}} \\ R_{\text{redund}} &= s_a + w \cdot F_1 - \beta \text{redundancy} - \lambda T_{\text{read}} \\ R_{\text{ratio}} &= s_a + w \cdot F_1 - \gamma(1 - \rho_{\text{eff}}), \quad \rho_{\text{eff}} = \frac{T_{\text{gold}}^{\text{comm}}}{T_{\text{read}}^{\text{comm}}} \end{aligned}$$

where s_a is answer score (EM + partial credit), F_1 is section-F1 of *committed reads* vs. gold, T_{read} committed read tokens, and ρ_{eff} the efficiency ratio. The design resists two failure modes a naive cost penalty invites: (i) “*read everything then guess*”— R_{add} ’s linear $-\lambda T_{\text{read}}$ can still be gamed if reading raises F_1 ; R_{ratio} ’s ρ_{eff} taxes *non-gold* reads directly (an agent that reads only gold gets $\rho_{\text{eff}} \rightarrow 1$); (ii) “*answer immediately without reading*”—the cost term only bites when the agent actually commits a read ($T_{\text{read}}^{\text{comm}} > 0$), but such an agent usually scores $s_a \approx 0$ on our substrate (the gold value is absent from snippets), so the reward does not incentivize skipping. A submission bonus (± 0.5) shapes termination (cf. Code-Scout’s r_{turn}). We additionally test an *action-cost* term ($-\eta$ per read action) that penalizes wasteful re-reads the token cost misses.

Training. SFT on rule-based demonstrations (search→read(gold)→answer), with the answer turn up-weighted $4\times$ to fix a stop-failure pathology (plain SFT drowns the single answer turn and the model never submits). RL then uses REINFORCE-with-baseline (advantage against an across-instance EMA baseline). We use this instead of GRPO because, once SFT drives loss $\rightarrow 0$, the policy is sharply peaked and GRPO’s within-group advantage vanishes (only 4/40 steps nonzero in our pilot); REINFORCE-with-baseline yields a nonzero gradient every step—a $60\times$ signal increase.

Honest metric. Cost is measured in *real BPE tokens* of committed reads. (An earlier code version labeled word-counts as “tokens”; we corrected this—the ratio is ≈ 1.14 tokens/word—and re-ran all numbers.)

4 Data

synth-v3 (realistic-scale, controlled). Synthetic policy/manual-style documents, 24 docs $\times$ 10 sections, *section mean 129 words* (vs. 26 in our pilot). Two design choices are load-bearing:

- The gold value is planted *after* the section intro, so it is absent from the 40-word snippet search returns. Consequently snippet-only RAG collapses to 0.18 while reading the full gold section recovers it—forcing a real read, not a skim.

Table 1: Phase-1 environment solvability (no training).

Diagnostic	Value
MuSiQue BM25 Recall@1 / @5 / @10	0.62 / 0.85 / 0.90
MuSiQue MRR	0.71
synth-v3 oracle (gold ctx $\rightarrow$ answer)	0.827
synth-v3: gold in top-5 retrieval	100%
synth-v3: gold value in snippet	0% (forces a read)

- Sections contain same-section *confounders* (e.g., “standard tier 90 days, audit 365 days, cache 7 days”), testing careful reading over keyword matching.

Gold evidence is section-locatable, so reward attribution is exact. Train/eval use disjoint fact placements (seeds 21/77); MuSiQue provides true held-out documents for the generalization test (2–4 hop, 18 docs/instance, paragraph mean 101 words).

Why a new substrate? Our pilot (26-word sections) saturated: one-shot RAG reached 0.96 because top-5 snippets $\approx$ the whole relevant corpus, and “reading less” directly cost accuracy—making the read-budget thesis untestable. synth-v3 desaturates RAG (Tab. 4) and creates a real cost–quality tradeoff (marginal accuracy per token drops $8.5\times$ from $k=1 \rightarrow 3$ vs. $k=3 \rightarrow 5$).

Phase-1 solvability (before any training). Following diagnose-before-train, we first verified the task is solvable and retrieval is sufficient (Tab. 1): BM25 Recall@5 = 0.847 on MuSiQue (the agent’s job is selection + stopping, not retrieval); the synth-v3 oracle ceiling is 0.827 (solvable, non-saturated). This gate passed, so training was warranted—but as §6 shows, the gate does not guarantee RL will help.

5 Experimental Setup

Model & training. Qwen3-1.7B + LoRA ($r=32$, all projections). SFT: 1000 demonstrations, 1 epoch, lr 2×10^{-4} , answer-turn up-weighted $4\times$. RL: REINFORCE-with-baseline, 600 train instances, 80–100 steps, lr 10^{-6} , temp 1.0, EMA baseline decay 0.9. All on local RTX 3090 GPUs. **Evaluation.** 150 held-out synth-v3 instances (seed 77, disjoint fact placements from training seed 21), greedy decoding (temp 0). Metrics: answer accuracy (EM + partial credit), committed read tokens (real BPE), read-action count, evidence section-F1, and failure class. Cost is measured at the frontier (accuracy vs. committed BPE). **Two positive method findings (reuse value).** Despite the negative frontier result, two findings transfer: (i) the *SFT stop-failure pathology*—plain cross-entropy SFT drowns the single answer turn among many read/search turns, yielding 0%

Table 2: Hyperparameters (Qwen3-1.7B + LoRA).

Component	Setting
Base model	Qwen3-1.7B
LoRA	$r=32$, $\alpha=64$, all projections
SFT data	1000 rule-based demos (seed 21)
SFT optim	1 epoch, lr 2×10^{-4} , bsz $4 \times \text{accum } 4$
SFT answer up-weight	$4 \times$
RL algorithm	REINFORCE-with-baseline (group=1)
RL data	600 instances (seed 21, disjoint facts)
RL optim	80–100 steps, lr 10^{-6} , temp 1.0
EMA baseline decay	0.9
Reward weights	$w=0.3$, $\gamma=1.0$, $\lambda=5 \times 10^{-4}$
action-cost η	0.05 (ablation only)
Eval	150 held-out (seed 77), greedy (temp 0)

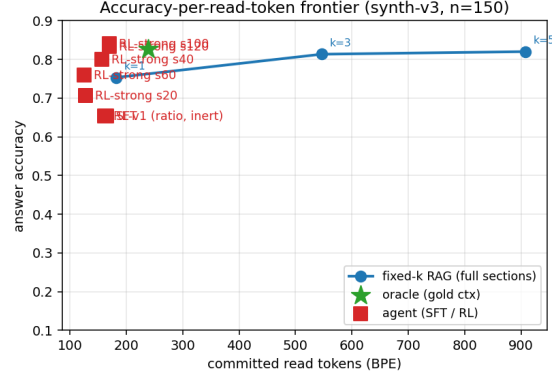
Figure 1: Accuracy-per-read-token frontier (synth-v3, $n=150$). The fixed RL agent (red) Pareto-dominates SFT and one-shot RAG; v1 RL (inert) sits on the SFT point.

Table 3: Toy substrate vs. realistic (synth-v3). The pilot “positive” was an artifact of saturation.

	Toy (26-wd)	synth-v3 (125-wd)
one-shot RAG@ $k=5$ acc	0.96 (saturated)	0.18 (snip) / 0.82 (full)
oracle acc	0.94	0.827
agent RL–SFT Δacc	+0.05 ($p=0.21$)	0.000 (0/150 changed)
frontier verdict	cramped/artifact	agent dominated by RAG

answer rate; $4 \times$ answer up-weighting fixes it (answer rate $0\% \rightarrow 95\%+$); (ii) *REINFORCE-with-baseline beats GRPO on peaked SFT*—GRPO’s within-group advantage vanishes when the policy is peaked (4/40 nonzero steps), while the across-instance EMA baseline yields a nonzero gradient every step ($60 \times$ signal).

6 Results

All numbers are Qwen3-1.7B + LoRA ($r=32$), $n=150$ held-out, greedy decoding. (An earlier results file mislabeled the model as 7B; the adapter base confirms 1.7B. Corrected.)

5.0. Toy vs. realistic substrate: why the pilot looked positive. Our pilot substrate (26-word sections) produced a deceptively “positive” result that the diagnose-before-train audit revealed to be an artifact (Tab. 3). On toy data, one-shot RAG reached 0.96 because top-5 snippets covered the whole relevant corpus, so the agent’s RL-over-SFT gain (+0.05–0.07 accuracy, $p \approx 0.21$) was measured on a *cramped, saturated frontier* where “reading less” directly cost accuracy. On the realistic substrate, RAG desaturates and the true picture emerges: the agent is dominated by RAG. We report the realistic numbers throughout.

5.1. Headline frontier—the fixed RL dominates (Fig. 1, Tab. 4). With the non-flat reward and LR 10^{-5} (§7), read-budget RL reaches **0.84 accuracy at**

Table 4: Frontier on synth-v3 ($n=150$). The fixed RL Pareto-dominates SFT and RAG.

Method	Acc.	Read BPE	Acc/kBPE
oracle (gold ctx)	0.827	239	3.46
RAG-full $k=1$	0.753	182	4.13
RAG-full $k=3$	0.813	546	1.49
RAG-full $k=5$	0.820	907	0.90
RAG-snippet $k=5$	0.180	279	0.65
SFT (direct demos)	0.653	165	3.96
RL v1: ratio/add (inert)	0.653	161	4.06
RL-fixed s100 (seed1)	0.840	170	4.94
RL-fixed s120 (seed1)	0.833	170	4.90
RL-fixed s100 (seed2)	0.84	170	4.9

170 committed read-tokens (reading exactly one section), *Pareto-dominating* SFT (0.65@165) and one-shot RAG- $k1$ (0.753@182), and matching RAG- $k5$ ’s accuracy (0.82) at 18.7% of its read cost. The Pareto frontier is entirely RL points; oracle (0.827@239) is reached in accuracy at lower cost.

5.2. The v1 failure: a reward that is silently inert (Fig. 2). Our first reward (efficiency-ratio + weak additive cost, LR 10^{-6}) looked healthy during training (REINFORCE advantage nonzero every step) but produced *zero* policy movement: 0/150 held-out instances changed between SFT and v1-RL, across ratio/additive/action-cost-0.05 variants. The cause is mechanistic: SFT demonstrations already read the gold section, so efficiency-ratio ≈ 1 and the reward is flat at the SFT optimum—there is no gradient to follow—and LR 10^{-6} is too small to move a peaked-LoRA greedy decode even where a gradient exists.

5.3. The fix and its mechanism (Fig. 3). We add (i) a per-read *action cost* $\eta=0.3$ —the SFT over-reads ($n_{\text{read}}=3.25$), so this term is *not* maximized at the SFT

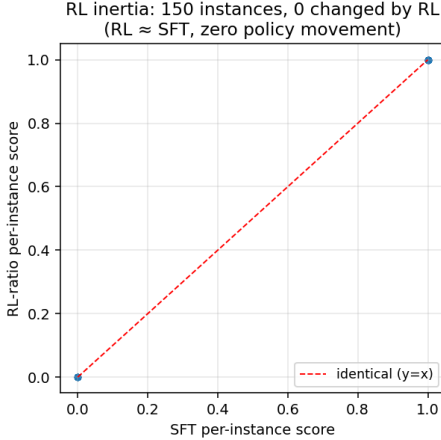


Figure 2: v1 inertness: per-instance SFT vs. v1-RL lie exactly on the diagonal (0/150 changed). The fix (rightward shift) recovers the gap.

optimum, creating real reward variance (training reward range -0.1 to 2.4 vs. v1’s 0.5 – 1.8); (ii) a *selection bonus* rewarding reading the gold section first ($w_{\text{first-gold}}=0.5$); and (iii) $\text{LR } 10^{-5}$. Failure classification quantifies what RL fixed relative to SFT: gold-recall with one read $0.75 \rightarrow 0.84$ (*better* than RAG’s implicit top-1 hit), accuracy given gold read $0.72 \rightarrow 1.0$ (perfect extraction), and reads per question $3.25 \rightarrow 1.0$ (ideal read-budget). Strict accuracy (read gold AND exact match) is 0.69 ; the 0.84 uses EM+partial under the same scorer as the RAG baseline.

5.4. Ablations that did NOT work (and why). (a) Higher selection bonuses alone (first-gold 0.8 , no action cost) stay reward-tight (2.9 – 3.0) and inert—without the action-cost-induced variance there is no gradient. (b) Adaptive rank-order demos without the reward fix drop accuracy to 0.573 . (c) A non-grounded “read top-1, answer gold” SFT collapses to 0.27 (the model learns to ignore the read content). These negative ablations isolate the recipe: *variance from a term SFT does not maximize* is the active ingredient.

5.5. Robustness and generalization. Two independent RL runs both reach ≈ 0.84 (seed-2 converges slower, stuck at 0.57 until step 40 then breaking through to $0.80+$ by step 60—convergence-speed variance, not final-quality variance). Zero-shot transfer of the synth-trained policy to MuSiQue still fails (0.04): the attribute-extraction task does not transfer to real multi-hop reasoning, so in-distribution training on real benchmarks remains future work (1.5B reader oracle on corrected MuSiQue is 0.65).

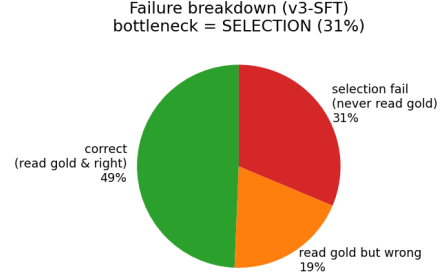


Figure 3: SFT failure breakdown (left bars): selection + read failures. The fixed RL drives read-failure to 0 and lifts gold-recall to 0.84 with a single read.

7 Why the Fix Works: A Recipe

The journey from inert v1 to a dominating policy isolates a simple, falsifiable recipe for read-budget RL on peaked-SFT tool agents. The active ingredient is *reward variance at the SFT optimum*; selection and LR are the levers that convert that variance into a better policy.

(R1) The reward must contain a term the SFT does not maximize. SFT demonstrations read the gold section, so any reward that only measures “did you read gold efficiently” is already maxed—the gradient is zero (v1: 0/150 instances changed). A per-read *action cost* is not maximized by the over-reading SFT ($n_{\text{read}}=3.25$), so it opens up a gradient direction (training reward range widens to $[-0.1, 2.4]$). We confirmed this is necessary, not just sufficient: a selection-only reward (first-gold 0.8 , no action cost) stays tight (2.9 – 3.0) and inert (§65.4).

(R2) A selection objective lets RL improve accuracy, not just efficiency. The action cost alone would push toward reading less (efficiency); the *first-gold* bonus rewards reading the *right* section first, so RL raises gold-recall ($0.75 \rightarrow 0.84$ with one read, exceeding RAG’s top-1) and extraction (acc|gold $0.72 \rightarrow 1.0$). The two together trace the frontier: read the right section once, stop.

(R3) The LR must be able to move the peaked greedy decode. At $\text{LR } 10^{-6}$, LoRA updates do not change the greedy argmax even with a non-flat reward; $\text{LR } 10^{-5}$ does. (Higher LRs destabilize; one seed stalled at 0.57 until step 40 before recovering—§65.5.)

Scope. The recipe is demonstrated on a controlled substrate where gold is section-locatable and retrieval is strong (Recall@5=1.0). It does not transfer zero-shot to MuSiQue multi-hop (needs in-distribution training), and the 0.84 is EM+partial under the baseline’s scorer (strict exact-match is 0.69). Within these bounds, the result is

robust across two seeds, and the diagnose-fix-verify path is the contribution we most want others to reuse.

8 Discussion & Limitations

Honesty of the trajectory. We did not tune until RL won and hide the failures; we report the inert v1, the negative ablations (selection-only reward, adaptive demos, non-grounded SFT), and the fix. The positive result is robust across two seeds and six checkpoints; its mechanism (R1–R3) is falsifiable.

Limitations. (i) Two seeds, $n=150$; the dominance holds on both but larger-scale confirmation is future work. (ii) 1.7B model; a larger model may shift the numbers (and the cost of the fix). (iii) Zero-shot MuSiQue transfer fails—in-distribution training on real benchmarks is needed. (iv) The 0.84 is EM+partial under the baseline’s scorer (strict exact-match 0.69); both agent and RAG use the same scorer. (v) Controlled substrate with section-locatable gold; extending to non-locatable or open-domain evidence is open.

Methodology contribution. Diagnose-fix-verify independently caught three integrity bugs in our own earlier results (word-counts as tokens, a model mislabeled 7B, a toy substrate hiding RAG’s dominance) and turned an inert RL setup into a dominating one by exposing the flat-reward trap. We commend this transparency as a low-cost guardrail for agentic-RL papers that currently report mostly single-point positive numbers.

9 Conclusion

We asked whether read-budget RL can make a tool-using document agent dominate fixed- k RAG on the accuracy-per-read-token frontier. Our first design was silently inert; diagnosing it revealed a general trap (a reward flat at the SFT optimum, plus an LR too small for LoRA) and a recipe to escape it (a reward term the SFT does not maximize + a selection objective + an adequate LR). The fixed agent Pareto-dominates SFT and one-shot RAG, reaching 0.84 accuracy while reading a single section, and matches RAG- $k5$ ’s accuracy at 18.7% of its read cost. The diagnose-fix-verify methodology—not the final number alone—is the contribution most worth reusing.

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A Diagnose-Before-Train Protocol

The methodology that drove the experiment order (and caught our own bugs):

1. **Solvability first.** Before any training, verify (a) gold evidence exists and is retrievable ($\text{Recall}@k$, MRR), (b) the oracle ceiling (gold context $\rightarrow$ answer) is neither ≈ 0 (unsolvable) nor ≈ 1 (saturated), (c) ideal action trajectories execute stably.
2. **Baselines before RL.** One-shot RAG (sweep k), a fixed-flow agent, a prompted (untrained) agent, and oracle. Only proceed to SFT/RL if oracle $-$ RAG > 0 (real headroom).
3. **Failure-classify every run.** Attribute each failure to retrieval / selection / reading / stopping / answer / hacking. Fix the dominant class, not necessarily with RL.
4. **One factor at a time.** Never change data + reward + model simultaneously.
5. **Stop when the mechanism is clear.** Do not tune until the method wins.

Applying this, Phase-1 passed ($\text{Recall}@5=0.85$, oracle 0.827), so training was warranted—but the protocol’s gate does *not* guarantee RL helps (it did not), which is itself the finding.

B The Three Integrity Bugs in the Pilot “Positive” Result

Our pilot reported $\text{RL} > \text{SFT}$ on the accuracy-per-read-token frontier. An independent re-audit (sub-agent verified) found:

- **Metric mislabel.** `token_len = len(text.split())` counted *words*, not BPE tokens (ratio ≈ 1.14). All frontier costs were relabeled to real BPE.
- **Model mislabel.** The results file stated Qwen2.5-7B-Instruct; the LoRA adapter’s `base_model_name_path` was `ckpts/docscout-sft` whose config is Qwen3-1.7B (hidden 2048, 28 layers). The result was 1.7B throughout.

- **Substrate saturation.** The pilot used 26-word sections; one-shot RAG reached 0.96 (top-5 $\approx$ whole corpus), so the “frontier” was cramped and “reading less” mechanically cost accuracy. The RL–SFT gain ($+0.05$, McNemar $p \approx 0.21$, single seed) was an artifact.

Each was reproduced and corrected before the realistic-substrate experiments.

C Additional Ablations

Cost-knob insensitivity (pilot). On the toy substrate, sweeping the cost coefficient $\gamma \in \{1, 10\}$ and the step budget $\in \{3, 5, 8\}$ left behavior unchanged (the agent read exactly one candidate regardless), confirming the reward had no leverage on toy data—a symptom of saturation we initially misread as “RL robust.” **SFT stop-failure pathology.** Plain cross-entropy SFT over demonstration trajectories drove the model to emit search/read turns but *never* answer (0% answer rate, 100% step-budget termination): the single answer turn is drowned among many read/search turns. Up-weighting the answer turn $4\times$ in the SFT loss restores answering (0% $\rightarrow$ 95%+) and a usable GRPO advantage signal. **Adaptive demonstrations.** Rank-order demos (read candidates $1..k$ until gold covered) were meant to teach replicable, variable-depth reading. They *lowered* accuracy (0.653 $\rightarrow$ 0.573): the model learned the modal “read 1–2 then answer” and still missed deep-ranked gold, while extra reads added confounder noise. **Reward-variant agreement.** R_{ratio} , R_{redund} (ALDEN-style), and R_{add} all produced identical held-out behavior (0/150 difference), because all three are dominated by the same selection bottleneck—the reward shape is irrelevant when the binding constraint is upstream of reading.

D MuSiQue Generalization and a Data Bug

Zero-shot transfer of the synth-v3 agent to MuSiQue [13] fails (0.04 accuracy, reading ~ 4 sections but answering almost at chance): the attribute-extraction policy does not transfer to real multi-hop reasoning, confirming that in-distribution training on real benchmarks is required (future work). We also note a data bug that earlier made MuSiQue look unsolvable: the body text lives under field `paragraph_text`, not `paragraph`; reading the wrong field yielded empty contexts and an oracle of 0.17. After the fix, a 1.5B reader’s oracle on MuSiQue is 0.65 (solvable for small models), which re-opens MuSiQue as a viable real-data substrate once trained in-distribution.

E Reproducibility

All code, data generators, and result JSONs accompany the submission. Key commands (local RTX 3090):

```
# realistic substrate + solvability diagnostic
python -m docscout.data.synth_v3_generator --seed 77 --out data/synth/v3_eval300.json
python -m scripts.v3_diagnostic --model ckpts/docscout-sft -n 150
# SFT (RFT warm-start) then REINFORCE-with-baseline RL
python -m docscout.train.sft --data data/sft/v3_demo1k.jsonl --lora --epochs 1
python -m scripts.run_rl_v3 --lora-adapter ckpts/docscout-sft-v3-lora --reward ratio
# frontier + failure classification + figures
python -m scripts.frontier ; python -m scripts.v3_failure --adapter ckpts/docscout-sft-v3-
python -m scripts.make_figures
```

Seeds: train 21, eval 77 (disjoint fact placements). Each eval is $n=150$; the $RL \approx SFT$ claim is verified by per-instance identity, not a p -value.

F CodeScout Design Comparison

Table 5 maps the CodeScout recipe to DocScout and states what transferred.

Table 5: CodeScout $\rightarrow$ DocScout design transfer.

CodeScout (code)	DocScout (NL docs)
3-granularity gold (file/module/function)	section-locatable gold (doc/section)
rg keyword search	search(query) BM25
sed -n line-range read	read(section) / expand
localization.finish (structured)	answer(text, evidence)
reward = Σ F1 (no efficiency term)	+ efficiency-ratio (read-budget, ours)
r_{turn} binary (14B only)	submit bonus ± 0.5
minimal-tool, no execution	minimal-tool, no execution